

SOG XIAN, China

WMO: 561060

Lat: 31.8877N

Lon: 93.7796E

Elev: 4024

StdP: 61.45

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.2	-17.0	-34.7	0.2	-3.7	-32.2	0.3	-4.6	11.2	2.1	10.1	0.9	0.5	290	0.686

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	21.2	9.9	19.8	9.7	18.5	9.3	12.0	17.3	11.2	16.9	10.5	16.2	2.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	13.1	13.9	9.3	12.1	13.1	8.5	11.4	12.7	48.7	17.2	45.9	17.1	43.6	16.3	16.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	7.7	6.4	DB	-22.8	24.0	2.7	1.3	-24.8	25.0	-26.4	25.8	-27.9	26.5	-29.8	27.5	(4)
(5)			WB	-23.3	12.9	2.4	1.1	-25.0	13.7	-26.4	14.4	-27.7	15.0	-29.4	15.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-7.2	-4.7	-1.0	3.2	7.0	10.8	12.8	12.5	9.7	3.9	-2.0	-6.1
	DBStd	7.57	3.64	3.41	2.91	2.43	2.62	2.44	1.79	1.97	2.11	3.01	2.70	3.47
	HDD10.0	2693	533	411	341	205	99	19	2	3	31	191	359	499
	HDD18.3	5493	792	644	599	455	351	226	170	181	260	448	609	758
	CDD10.0	242	0	0	0	0	7	43	90	81	21	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	3	0	0	0	0	0	1	1	2	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	1.9	2.1	2.3	2.5	2.4	2.1	1.8	1.6	1.7	1.6	1.8	1.7	1.8
Precipitation	PrecAvg	569	6	5	8	15	49	130	129	105	94	26	5	3
	PrecMax	702	22	16	21	39	94	196	214	221	156	81	16	12
	PrecMin	438	0	0	1	1	6	39	40	28	29	2	0	0
	PrecStd	83	6	4	5	10	23	39	40	42	30	17	5	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	8.9	13.5	16.3	20.4	22.4	22.6	23.2	20.0	17.0	10.4	8.4
		MCWB	-1.4	-1.7	0.2	1.8	6.3	9.4	11.0	9.7	9.3	5.0	-1.0	-2.2
	2%	DB	5.6	6.5	10.1	13.5	17.4	19.5	20.9	21.3	18.0	14.0	8.3	5.7
		MCWB	-3.2	-3.0	-0.7	1.4	5.5	9.5	10.5	9.9	8.8	4.4	-1.5	-3.4
	5%	DB	2.8	4.6	7.9	11.5	15.3	17.8	19.5	19.8	16.5	11.9	6.6	3.6
		MCWB	-4.9	-4.0	-1.7	0.8	4.5	8.9	10.0	10.0	8.2	3.4	-2.2	-4.5
	10%	DB	0.5	2.7	6.0	9.9	13.4	16.2	18.0	18.1	14.9	9.9	4.8	1.5
		MCWB	-6.1	-4.8	-2.6	0.3	4.1	8.4	9.7	9.7	7.8	2.6	-2.9	-5.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.1	-1.0	1.5	4.3	8.7	11.9	13.4	13.6	11.2	7.3	0.3	-1.4
		MCDB	7.2	7.2	10.1	10.1	16.1	17.5	18.6	18.2	16.9	12.5	6.5	7.1
	2%	WB	-2.9	-2.6	0.0	3.0	7.4	10.7	12.1	12.2	10.1	6.1	-0.7	-3.1
		MCDB	4.4	4.8	8.0	9.2	14.2	16.8	17.7	16.7	15.5	11.9	5.8	4.4
	5%	WB	-4.6	-3.5	-1.0	2.0	6.2	9.9	11.3	11.2	9.2	5.0	-1.5	-4.3
		MCDB	1.8	3.3	6.2	9.0	12.3	16.0	17.2	16.5	14.5	9.7	4.9	2.6
	10%	WB	-5.9	-4.4	-1.9	1.1	5.3	9.2	10.6	10.4	8.5	3.8	-2.3	-5.3
		MCDB	-0.3	1.8	4.6	7.9	10.7	14.6	16.3	15.9	13.6	8.0	3.8	0.9
Mean Daily Temperature Range	5% DB	MDBR	14.0	13.0	12.4	11.9	10.8	9.9	10.3	10.9	10.8	10.9	13.5	14.9
		MCDBR	18.4	16.7	15.8	15.7	14.8	12.7	13.0	14.3	13.4	14.4	16.8	18.4
		MCWBR	11.4	9.9	8.1	7.0	5.6	4.7	4.7	4.9	5.4	6.1	8.7	11.3
	5% WB	MCDBR	16.9	15.2	13.4	12.4	10.9	10.6	10.5	10.5	11.1	11.1	13.8	17.3
		MCWBR	10.8	9.3	7.2	5.8	4.9	4.7	4.5	4.5	5.0	5.1	7.4	11.1
Clear-Sky Solar Irradiance	taub		0.189	0.208	0.252	0.290	0.297	0.286	0.281	0.276	0.259	0.231	0.195	0.179
	taud		2.648	2.578	2.454	2.387	2.447	2.583	2.654	2.654	2.672	2.674	2.699	2.738
	Ebn at Noon		1085	1079	1038	1000	987	992	996	1002	1017	1037	1069	1089
	Edh at Noon		72	87	107	120	114	99	92	91	85	78	68	62
All-Sky Solar Radiation	RadAvg		3.55	4.37	5.31	5.92	6.11	5.88	5.86	5.51	4.92	4.38	4.02	3.48
	RadStd		0.34	0.25	0.35	0.36	0.39	0.35	0.36	0.40	0.25	0.28	0.19	0.24

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.59	N/A	N/A	N/A	-0.65	-0.70	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.51	+0.77	-1.95	N/A	N/A	N/A	-168	-218	+42

Nomenclature: See separate page